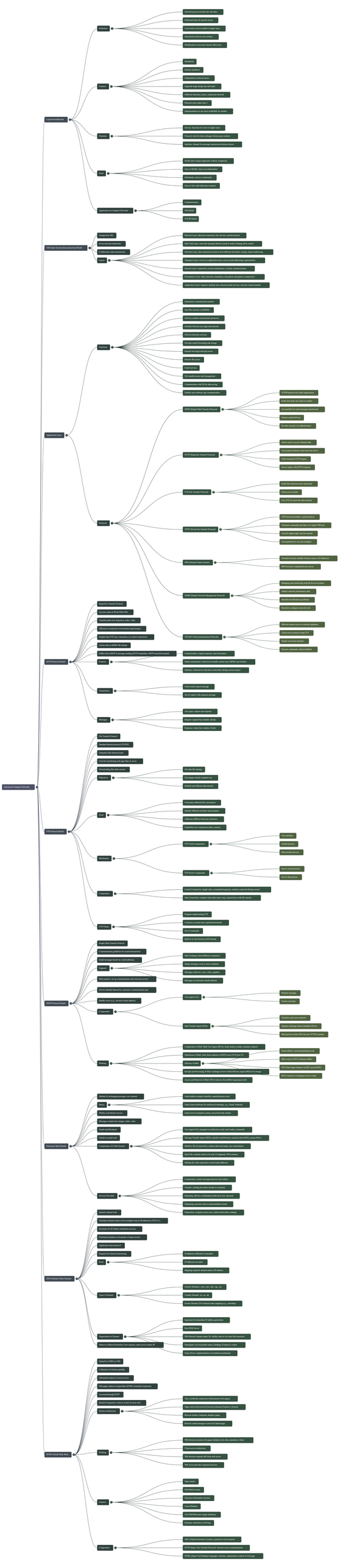

- Advanced Computer Networks
  - Layered Architecture
    - Definition
      - Network process divided into sub-tasks
      - Dedicated tasks for specific layers
      - Lower layer services added to higher layer
      - Interactions between sub-systems
      - Modification in one layer doesn't affect next
    - Features
      - Modularity
      - Distinct interfaces
      - Independence between layers
      - Segments large design into sub-tasks
      - Different functions, layers, content per network
      - Physical route under layer 1
      - Implementation in one layer modifiable by another
    - Elements
      - Service: functions by lower to higher layer
      - Protocol: rules for data exchange between peer entities
      - Interface: channel for message transmission between layers
    - Need
      - Divide and Conquer Approach: reduces complexity
      - Easy to Modify: layers are independent
      - Modularity: easier to understand
      - Easy to Test: individual layer analysis
    - Application in Computer Networks
      - Communication
      - OSI Model
      - TCP/IP Model
  - OSI (Open System Interconnection) Model
    - Designed by ISO
    - Seven layered architecture
    - Collaborative data transmission
    - Layers
      - Physical Layer: physical connection, bits, bit rate, synchronization
      - Data Link Layer: error-free message delivery (node to node), framing, flow control
      - Network Layer: data transmission (host to host different networks), routing, logical addressing
      - Transport Layer: services to application layer, service point addressing, segmentation
      - Session Layer: connection, session maintenance, security, synchronization
      - Presentation Layer: data extraction, translation, encryption, decryption, compression
      - Application Layer: topmost, desktop layer, directory/mail services, network virtual terminal
  - Application Layer
    - Functions
      - Determines communication partner
      - Specifies resource availability
      - Delivers seamless transmission protocols
      - Interface between user apps and network
      - Delivers directory services
      - Provides email forwarding and storage
      - Remote host login and app access
      - Remote file access
      - Email services
      - File transfer access and management
      - Communicates with OS for data saving
      - Enables user-software app communication
    - Protocols
      - SMTP (Simple Mail Transfer Protocol)
        - TCP/IP protocol for email organization
        - Sends data from one email to another
        - Accountable for email message transmission
        - Ensures email delivery
        - Provides security via authentication
      - HTTP (Hypertext Transfer Protocol)
        - Allows users to access Internet data
        - Conversation between client and web server
        - Client transmits HTTP request
        - Server replies with HTTP response
      - FTP (File Transfer Protocol)
        - Sends files between server and client
        - Client-server model
        - Uses TCP for error-free data transfer
      - TFTP (Trivial File Transfer Protocol)
        - UDP-based (unreliable, connectionless)
        - Transmits commands and data over single UDP port
        - Used for lightweight, fast file transfer
        - Uncomplicated to use and configure
      - DNS (Domain Name System)
        - Translates human-readable domain names to IP addresses
        - Web browsers comprehend user access
      - SNMP (Simple Network Management Protocol)
        - Managing and monitoring network devices/systems
        - Gathers network performance data
        - Identifies/troubleshoots problems
        - Remotely configures network tools
      - TELNET (Telecommunication Network)
        - Delivers remote access to network appliance
        - Client-server protocol using TCP
        - Simple text-based interface
        - Execute commands, obtain feedback
  - HTTP Protocol Details
    - HyperText Transfer Protocol
    - Accesses data on World Wide Web
    - Transfers plain text, hypertext, audio, video
    - Efficiency in hypertext environment (rapid jumps)
    - Simpler than FTP (one connection, no control connection)
    - Carries data in MIME-like format
    - Differs from SMTP in message sending (HTTP immediate, SMTP stored/forwarded)
    - Features
      - Connectionless: request-response, then disconnect
      - Media independent: client/server handle content type (MIME-type header)
      - Stateless: client/server only know each other during current request
    - Transactions
      - Client sends request message
      - Server replies with response message
    - Messages
      - Two types: request and response
      - Request: request line, headers, (body)
      - Response: status line, headers, (body)
  - FTP Protocol Details
    - File Transfer Protocol
    - Standard Internet protocol (TCP/IP)
    - Transmits files between hosts
    - Used for transferring web page files to server
    - Downloading files from servers
    - Objectives
      - Provides file sharing
      - Encourages remote computer use
      - Reliable and efficient data transfer
    - Need
      - Overcomes different file conventions
      - Handles different text/data representation
      - Addresses different directory structures
      - Establishes two connections (data, control)
    - Mechanism
      - FTP Client Components
        - User interface
        - Control process
        - Data transfer process
      - FTP Server Components
        - Server control process
        - Server data process
    - Connections
      - Control Connection: simple rules, commands/responses, remains connected during session
      - Data Connection: complex rules (data types vary), opens/closes with file transfer
    - FTP Clients
      - Program implementing FTP
      - Connects to remote host, uploads/downloads
      - Set of commands
      - Built-in in web browsers (GUI-based)
  - SMTP Protocol Details
    - Simple Mail Transfer Protocol
    - Communication guidelines for email transmission
    - Sends messages based on e-mail addresses
    - Supports
      - Mail exchange (same/different computers)
      - Single message to one or more recipients
      - Messages with text, voice, video, graphics
      - Messages on networks outside Internet
    - Main purpose: set up communication rules between servers
    - Servers identify themselves, announce communication type
    - Handle errors (e.g., incorrect email address)
    - Components
      - User Agent (UA)
        - Prepares message
        - Creates envelope
      - Mail Transfer Agent (MTA)
        - Transfers mail across Internet
        - Supports relaying system (multiple MTAs)
        - Mail gateway (relay MTA for non-TCP/IP systems)
    - Working
      - Composition of Mail: Mail User Agent (MUA), body, header (sender, recipient, subject)
      - Submission of Mail: mail client submits to SMTP server (TCP port 25)
      - Delivery of Mail
        - Email address: username@domain.com
        - MSA sends to MTA if domain differs
        - MTA finds target domain via MX record (DNS)
        - MTA connects to exchange server to relay
      - Receipt and Processing of Mail: exchange server to Mail Delivery Agent (MDA) for storage
      - Access and Retrieval of Mail: MUA retrieves from MDA (login/password)
  - Electronic Mail (Email)
    - Method of exchanging messages over internet
    - Basics
      - Email address (unique identifier, name@domain.com)
      - Email client (software for send/receive/manage, e.g., Gmail, Outlook)
      - Email server (computer system, stores/forwards emails)
    - Widely used Internet service
    - Messages contain text, images, audio, video
    - Sender and Recipient
    - Similar to postal mail
    - Components of E-Mail System
      - User Agent (UA): program to send/receive mail, mail reader, commands
      - Message Transfer Agent (MTA): transfers mail between systems (client MTA, system MTA)
      - Mailbox: file on local drive, collects delivered mails, user reads/deletes
      - Spool file: contains mails to be sent, UA appends, MTA extracts
      - Mailing list: alias represents several email addresses
    - Services Provided
      - Composition: creates messages/answers (text editor)
      - Transfer: sending procedure (sender to recipient)
      - Reporting: delivery confirmation (delivered, lost, rejected)
      - Displaying: presents mail in understandable format
      - Disposition: recipient action (save, delete before/after reading)
  - DNS (Domain Name System)
    - Internet's phone book
    - Translates domain names (www.example.com) to IP addresses (192.0.2.1)
    - Hostname for IP address translation services
    - Distributed database in hierarchy of name servers
    - Application layer protocol
    - Required for Internet functioning
    - Need
      - IP addresses difficult to remember
      - IP addresses not static
      - Mapping required: domain name to IP address
    - Types of Domain
      - Generic Domains: .com, .edu, .mil, .org, .net
      - Country Domain: .in, .us, .uk
      - Inverse Domain: IP to domain name mapping (e.g., nslookup)
    - Organization of Domain
      - Important for immediate IP address generation
      - Root DNS Server
      - DNS Record: domain name, IP, validity, time to live (tree-like structure)
      - Namespace: set of possible names, bindings of names to values
      - Name Server: implementation of resolution mechanism
    - Name-to-Address Resolution: host requests, name server returns IP
  - WWW (World Wide Web)
    - Started by CERN in 1989
    - Collection of websites globally
    - Information shared via local servers
    - Web pages linked via hyperlinks (HTML-formatted, hypertext)
    - Accessed through HTTP
    - Benefit of hypertext: click on words for more info
    - System Architecture
      - Vast, worldwide connection of documents (web pages)
      - Pages retrieved/viewed by browsers (Internet Explorer, Chrome)
      - Browser fetches, interprets, displays pages
      - Browser sends message to server for linked page
    - Working
      - Web browser accesses web pages (displays text, data, animation, video)
      - Client-server architecture
      - Web browser requests info from web server
      - Web server provides requested services
    - Features
      - Open source
      - Distributed system
      - Hypertext Information System
      - Cross-Platform
      - Uses Web Browsers (single interface)
      - Dynamic, Interactive, Evolving
    - Components
      - URL (Uniform Resource Locator): system for web resources
      - HTTP (Hyper Text Transfer Protocol): browser-server communication
      - HTML (Hyper Text Markup Language): structure, organization, content of web page